

Standard Operating Procedure (SOP) & Technical Manual

Enterprise IT Infrastructure, CCTV Surveillance & SLA Management

Comprehensive operational guidelines for sales inquiry processing, technical site surveys, structured cabling execution, BOQ formulation, and mini-CRM lifecycle management across Vapi, Silvassa, and Daman.

DOCUMENT CODE

SOP-NG-IT-001 (Rev 1.0)

TARGET REGIONS

Vapi GIDC • Silvassa • Daman • Sarigam • Umbergaon

AUTHORIZED OEM PARTNERS

Cisco • Hikvision • Sophos • Dell • HPE • APC Schneider

EFFECTIVE DATE

September 2026

OPERATIONAL SCOPE

Sales, Network Engineering, CCTV, IT AMC, Web CRM

CLASSIFICATION

Internal Operations & Client SLA Compliance

1. Company Profile & Regional Mission

NextGen IT Solution is a specialized technology infrastructure and surveillance engineering company headquartered in Vapi, Gujarat. We bridge the gap between complex enterprise IT standards and the rugged demands of heavy industrial plants, pharmaceutical facilities, packaging mills, and corporate offices across South Gujarat and the Union Territories.

Core Regional Value Proposition

Unlike generic metropolitan IT vendors or retail computer shops, NextGen IT Solution maintains local warehousing of critical networking components, fiber optic fusion splicers, and a dedicated roster of field engineers providing **2 to 4 hour emergency onsite response SLAs** in Vapi GIDC, Silvassa (Piparia/AmlI), and Daman (Kachigam/Somnath).

1.1 Regional Industrial Coverage

Industrial Zone	Key Estates Covered	Primary Industries	Standard SLA
Vapi GIDC	Phases I–IV, Morai, Chanod, Balitha	Chemicals, Dyes, Pharma, Paper Mills	2 - 3 HOURS
Silvassa (DNH)	Piparia, AmlI, Masat, Rakholi, Sayli	Textiles, Yarn, Plastics, FMCG	3 - 4 HOURS
Daman (DD)	Kachigam, Somnath, Dabhel, Ringanwada	Pharma formulations, Packaging, Molding	3 - 4 HOURS
Sarigam & Umbergaon	Sarigam GIDC, Umbergaon Industrial Area	Chemical Processing, Engineering Units	4 HOURS

1.2 Core Technology Architecture (15 Offerings)

- Structured Cabling & LAN:** TIA/EIA compliant CAT6/CAT6A copper networks, 24U–42U server rack dressing, patch panels, and fluke certification.
- CCTV & Video Surveillance:** 4K AcuSense IP cameras, DarkFighter low-light color, AI human/vehicle perimeter detection, and NVR arrays.
- Fiber Optic Backbone:** Single-mode (OS2) and multi-mode (OM3/OM4) armored optical fiber laying, fusion splicing, and OTDR testing.
- Firewall & Cybersecurity:** Next-Gen Firewalls (Sophos XGS, Fortinet), multi-WAN failover, Site-to-Site IPsec VPNs, and ransomware defense.
- Servers & Storage:** Dell PowerEdge & HPE rack servers, Synology enterprise NAS, hardware RAID (1/5/6/10), and virtualization.
- IT AMC & Onsite SLA:** Comprehensive & Non-Comprehensive maintenance contracts for computers, switches, and servers.
- Commercial Hardware:** CAD/CAM workstations, Dell/HP desktops, corporate laptops, and thermal barcode printers.

2. Standard Operating Procedure: Sales & Inquiries

This SOP governs how all prospective client inquiries, web form submissions, and tender Bill of Quantities (BOQ) uploads are recorded, evaluated, and transitioned into closed projects.

1 Inquiry Capture & Lead Ingestion

Inquiries arrive automatically via the web portal (/quote or /contact) or direct WhatsApp/Phone. Every submission automatically creates a unique record in the database with status **NEW** (e.g. LEAD-1005).

2 CRM Verification & BOQ Download

The sales desk opens the Mini-CRM dashboard at /admin/leads. The coordinator verifies client details (Company Name, Location, Phone) and downloads the attached BOQ file (PDF/Excel) from the "View BOQ" link.

3 First Response Protocol (SLA < 2 Hours)

The coordinator clicks the direct **WhatsApp** button on the lead row or calls the client. Confirm project scope, ask clarifying questions (number of CCTV points, ceiling height, building distances), and update status to **CONTACTED**.

4 Site Survey & Technical Costing

If the project requires physical verification, dispatch a field engineer within 24 hours. The engineering team applies authorized B2B vendor pricing (Cisco, Hikvision, Sophos, Dell) and formats an official PDF quotation.

5 Quotation Dispatch & Pipeline Tracking

Send quotation via email and WhatsApp. In the CRM, change lead status to **QUOTATION SENT** and record the quotation number in the internal notes field.

6 Deal Closing & Handover

Upon receiving Purchase Order (PO) or advance payment, update status to **WON**. If the client defers or cancels, update status to **Lost** with explanatory notes.

2.1 Pipeline Status Lifecycle Matrix

Status	Meaning	Action Required	Target Turnaround
NEW	Uncontacted inquiry received	Review BOQ, call or WhatsApp client	Within 2 Hours
CONTACTED	Initial discussion completed	Schedule site visit or calculate BOQ	Within 24 Hours
QUOTATION SENT	Formal proposal delivered	Follow up with purchase committee	Every 3 Days
NEGOTIATION	Commercial terms being finalized	Finalize volume discounts & delivery terms	Within 5 Days
WON	Deal closed, PO issued	Handover to Field Engineering Team	Immediate

3. SOP: Technical Field Engineering Operations

This SOP provides compulsory technical standards for cabling, CCTV deployment, server configuration, and quality certification on client sites.

3.1 Physical Site Survey Checklist

Check Item	Technical Standard	Risk Mitigation
Cable Pathway Distances	Copper CAT6 runs must NOT exceed 90 meters from switch to IO.	If >90m, mandate fiber optic link or intermediary PoE switch.
Electromagnetic Interference	Keep data cables >300mm away from 415V 3-phase industrial power cables.	Use shielded CAT6A STP or armored fiber in machine rooms.
Electrical Earthing	Server rack & switch earthing must measure <2.0 Ohms ground resistance.	Prevents equipment damage from industrial surges or lightning.
CCTV Lens Selection	Use 2.8mm for wide indoor rooms; 4mm/6mm for factory shop floor corridors; 12mm+ for entry gate number plates.	Eliminates blind spots and provides crisp evidence.

3.2 Certified Installation & Rack Dressing Standards

- Conduit Protection:** All cables must run inside heavy-duty rigid PVC or galvanized iron (GI) conduits with metal saddles every 1 meter. No hanging or loose cables permitted.
- Printed Ferrule Labeling:** Every single cable must have machine-printed heat-shrink ferrule tags at both ends (e.g. RACK1-SW2-PORT14). Hand-written tape is strictly prohibited.
- Rack Cable Management:** Use 1U brush panels and horizontal cable managers. Bundle patch cords with reusable hook-and-loop velcro straps (never plastic zip ties).
- Optical Fiber LIU:** Fusion-spliced pigtails must be neatly coiled inside the Light Interface Unit (LIU) tray with minimum bend radius >30mm.
- Testing & Certification:** 100% of terminated network points must be tested using a calibrated Fluke Cable Analyzer. Export PDF wiremap & throughput certificates for the client handover binder.

3.3 CCTV Quality Assurance Protocol

- Verify weatherproof IP67 cable gland seals on all outdoor dome and bullet cameras.
- Set camera encoding to **H.265+** smart compression to maximize recording retention (30–90 days).
- Configure AI intrusion detection lines across boundary walls and calibrate sensitivity to filter out stray animals and rain.
- Set up secure remote viewing credentials with strong alphanumeric passwords and role-based permissions (Security Guard view vs Director playback).

4. SOP: Annual Maintenance Contracts & System Ops

4.1 AMC Contract Tiers & Response Matrix

Contract Tier	Target Client	Scope of Support	Emergency Onsite SLA
Tier 1: Non-Comprehensive	General corporate offices, commercial shops	Preventive servicing, OS patches, labor only (client pays for parts)	4 to 6 Business Hours
Tier 2: Comprehensive	Continuous manufacturing plants, GIDC units	All hardware replacement spares included, standby loaner units	2 TO 4 HOURS
Tier 3: Facility Management (FMS)	Large chemical complexes, pharma campuses	Dedicated full-time resident engineer stationed on-site	IMMEDIATE (0 MIN)

4.2 Monthly Preventive Maintenance Schedule

Week 1: Server & Backup Audit Check RAID array health, verify automated cloud & NAS backups, inspect server event logs, and verify UPS battery runtime.
Week 2: Network & Firewall Audit Inspect managed switch port errors, check firewall threat logs, verify VPN connections, and clean network rack dust filters.
Week 3: CCTV & Surveillance Audit Clean camera optical lenses, verify HDD health in NVR, confirm continuous recording on all channels, and adjust night IR illumination.
Week 4: Desktop & Workstation Health Perform internal dust blowing, test thermal paste on CPUs, verify antivirus definitions, and clear temporary files.

4.3 Web Portal & Mini-CRM Administration Commands

Action	Command / Procedure
Start Dev Server	cd C:\Users\yatin.patel\.gemini\antigravity\scratch\nextgen-it-solutions && npm run dev
Build for Production	npm run build && npm run start
Access Mini-CRM	Navigate to http://localhost:3000/admin/leads
Lead Backup	Backup data/leads.json and public/uploads/ to external drive weekly

Document Sign-off & Maintenance

This Standard Operating Procedure is maintained by NextGen IT Solution Engineering Operations. For updates, suggestions, or escalation queries, contact: support@nextgenitsolution.com | **Hotline: +91 98251 12345.**